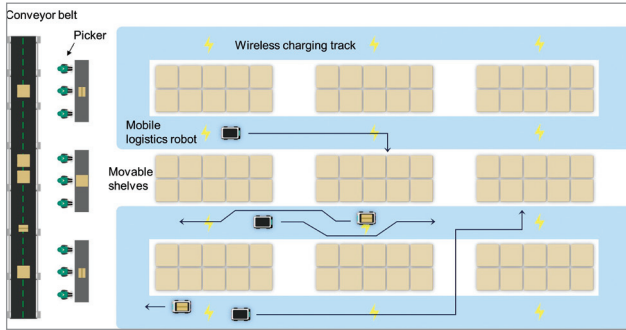


Wireless charging powers vehicles in motion

Advancements in electric vehicle (EV) technology are minimising the need for charging stations. A research team from UNIST has developed a wireless charging system that powers vehicles in motion via a robust magnetic field



Application of a track type dynamic wireless power transfer system in logistics (Credit: Applied Energy, 2024)

generated by a wired network. This eliminates costly ferromagnetic materials and allows vehicles to move freely along power tracks. The system improves power transfer efficiency by up to 90% with optimised power track and receiver algorithms. Safety standards from IEEE and ICNIRP are met. This technology could reduce battery size, decrease charging times, and lower pollution linked to lithium extraction, promoting sustainability.

Portable system harvesting water from air

An eco-friendly technology for water harvesting has been developed in South Korea, inspired by natural mechanisms in plants and animals. Researchers at the Korea Institute of Machinery and Materials (KIMM) created a portable system that captures and purifies moisture from the air, producing drinkable water. Weighing only three kilograms, the system has received safety certifications and is highly energy-efficient, doubling the performance of traditional dehumidifiers. It uses thermoelectric modules and sterilises bacteria with heat.

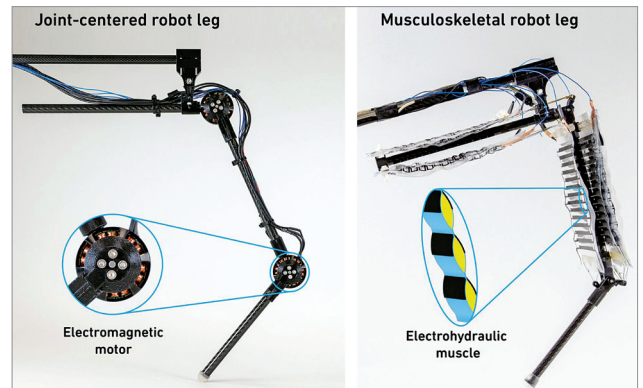
Optical computing revolutionises DOA estimation efficiency

Wireless sensing and communication are crucial in civilian and military applications, with direction-of-arrival (DOA) estimation playing a key role. Traditional methods like the MUSIC algorithm rely on complex RF circuitry, leading to high power consumption and latency. To overcome these limitations, optical computing offers a promising

alternative. Researchers at Tsinghua University developed super-resolution diffractive neural networks (S-DNNs) for all-optical DOA estimation. These surpass the Rayleigh diffraction limit, providing faster, more accurate results without RF circuits. S-DNNs operate by modulating electromagnetic waves, improving performance and robustness, especially in reconfigurable intelligent surfaces (RIS) for 6G communications.

Artificial muscle powered robotic leg mimics biological leg

A team from ETH Zurich and the Max Planck Institute have introduced a muscle-powered robotic leg using electrohydraulic actuators, called HASELs, which mimic biological muscle movements. These actuators are energy-efficient,



While conventional robotic legs are driven by an electromagnetic rotary motor (left), for their musculoskeletal system the researchers use electrohydraulic actuators, i.e. artificial muscles (right) (Credit: Thomas Buchner/ETH Zurich and Toshihiko Fukushima/MPI-IS)

avoiding the heat waste of traditional motors. The robotic leg can perform high jumps and adapt to uneven terrain with simple input signals, offering agility and efficiency. This innovation could transform future robots, especially in rescue missions or delicate tasks, by enhancing mobility and adaptability.

Novel hot-emitter transistor for post-Moore era

Researchers are advancing transistor technology to address the limitations of shrinking transistor sizes. A team from the Institute of Metal Research at the Chinese Academy of Sciences developed a novel hot-emitter transistor (HOET) using a new mechanism called stimulated emission of heated carriers (SEHC). This transistor, combining graphene with germanium, achieves an ultralow sub-threshold swing of less than 1mV/dec and a peak-to-valley current ratio above 100. These features enable faster, more energy-efficient devices, suitable for post-Moore era applications.